CHE 321 HW 10 Alvy Villaturo

1)

$E_A = 100 \frac{kJ}{mol}$ $T = 400K$ $W = 500$ g catalyst

$x = 0.50$

$\rho_c = 2.5 \; g/cm^3$ $e = 0.4$

a) $\frac{dF_A}{dV} = -\eta k'' S_a \rho_c (1-\varepsilon) C_A$ $F_A = C_A v$

$$\frac{dC_A}{dV} = \frac{-\eta k'' S_A \rho_c (1-\varepsilon) C_A}{v}$$

$$\int \frac{dC_A}{C_A} = \int \frac{-\eta k'' S_A \rho_c (1-\varepsilon)}{v} dV \qquad A = \frac{\eta k'' S_A \rho_c (1-\varepsilon)}{v}$$

$$\ln\left(\frac{C_A}{C_{A_0}}\right) = -A W (1-\varepsilon) \rho_c$$

$$\ln\left(\frac{\frac{1}{2} C_{A_0}}{C_{A_0}}\right) = -A W (1-\varepsilon) \rho_c$$

$$\frac{-\ln\left(\frac{1}{2}\right)}{W(1-\varepsilon)\rho_c} = A \qquad A = 0.000924$$

b) $k_1'' = A e^{\frac{-E_a}{RT}}$ $\qquad 1 \cdot 10^5 \frac{\cancel{cm}}{\cancel{min}} \cdot \frac{1m}{100\,\cancel{cm}} \cdot \frac{1\,\cancel{min}}{60\,s} = 11.67$

$k_1'' = 11.67 e^{\frac{-100,000}{8.314 \cdot 460}}$ $\qquad \frac{100\,\cancel{kJ}}{mol} \cdot \frac{1000\,J}{1\,\cancel{kJ}} = \frac{100,000\,J}{mol}$

$k_1'' = 1.45 \cdot 10^{-12}\ \frac{m}{s}$

$k_2'' = k_1 e^{\frac{-E_A}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)}$

$= 8.53 \cdot 10^{-12}\ \frac{m}{s}$

$A_2 = A_1 \frac{k_2''(k_1''^{\frac{1}{2}})}{k_1''(k_2^{\frac{1}{2}})}$

$A_2 = 0.002237505$

$\frac{-\ln\left(\frac{1}{2}\right)}{A(1-\varepsilon) r_2} = W$ $\qquad W = 201\ g$ catalyst

2) $A \rightarrow B$ $\quad -r_{As}' = k' C_{As}$ $\quad T = 300K$

$v_0 = \frac{10{,}000\ cm^3}{s}$ $\quad Sh = 100\, Re^{\frac{1}{2}}$ $\quad x = 0.60$

$C_{A0} = 1.0 \frac{mol}{L}$

$Re = \frac{9 \frac{\cancel{cm}}{\cancel{s}} \cdot 15 \cancel{cm}}{0.02 \frac{cm^2}{s}} = 67.5$ $\quad Sh = 821.6$

$Sh = \frac{k_c d_p}{D_{AB}}$ $\quad \frac{Sh\, D_{AB}}{d_p} = k_c$ $\quad k_c = 54.77 \frac{cm}{s}$

$\frac{\frac{cm^2}{s}}{\cancel{cm}}$

$k_{eff} = \frac{k_c k''}{(k_c + k'')}$ $\quad k'' = \frac{k'}{a} = \frac{.01 \frac{cm^{\cancel{3}}}{s g}}{60 \frac{cm^{\cancel{2}}}{g}} = 0.0001666667 \frac{cm}{s}$

$k_{eff} = 0.00017 \cdot 10^{-4} \frac{cm}{s}$

$\frac{dF_A}{dW} = k_{eff} C_A a$

$\ln\left(\frac{C_A}{C_{A0}}\right) = \frac{-W}{v_0} k_{eff} a$ $\quad C_A = (1-x) C_{A0} = \frac{.4 \cancel{C_{A0}}}{\cancel{C_{A0}}}$

$\frac{-v_0 \ln\left(\frac{C_A}{C_{A0}}\right)}{k_{eff}\, a}$ $\quad W = 9.16 \cdot 10^5\, g$

$\frac{\frac{cm^3}{s}}{\frac{cm}{s} \frac{cm^2}{g}}$